

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 4B

Distributions and relationships: calculate, draw and explain.

Bellerbys
Preparation for University Education

EXERCISE B0

Class A revision

Slides 3-4

1. Which interval includes 40: $30 \leq x < 40$ or $40 \leq x < 50$? Why?

2. Midpoint of $30 \leq x < 40$:

3. Seven of 25 sessions: relative frequency = _____; percentage = _____.

EXERCISE B1

Build a histogram from scratch

Slides 10-12

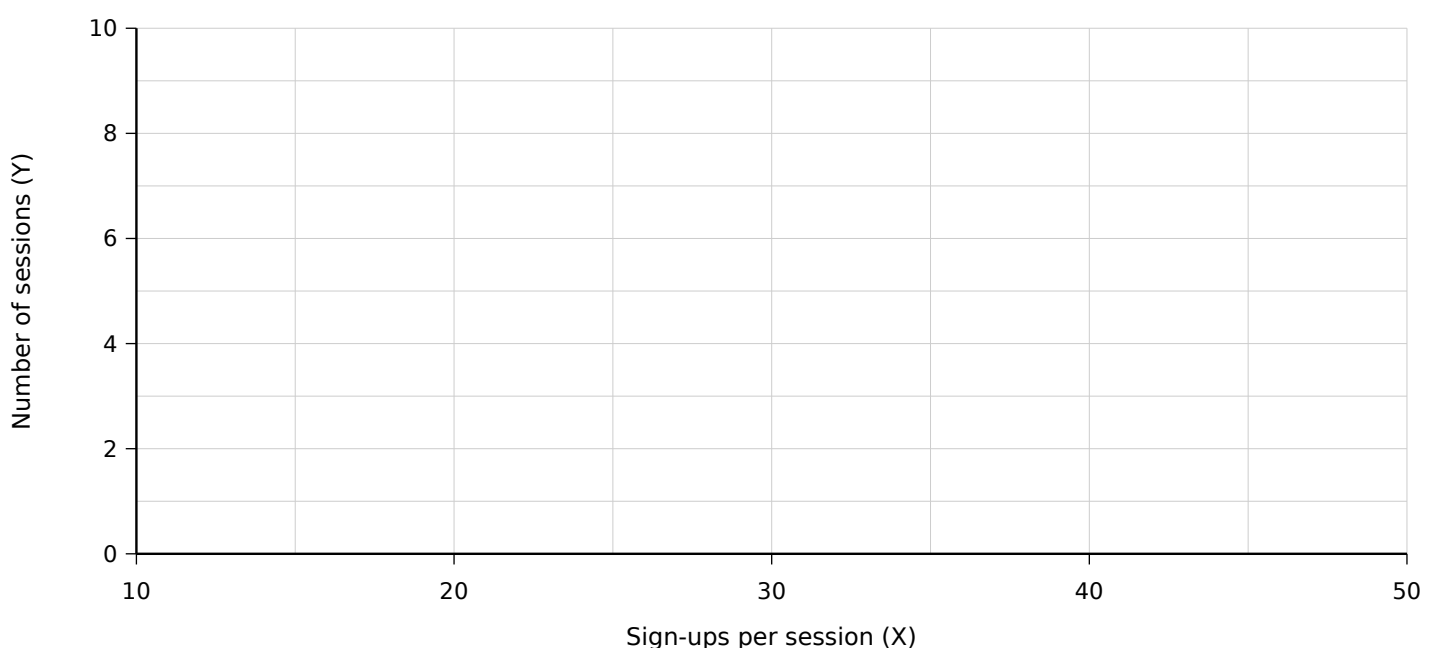
Smoothie Club: each value is the sign-ups in one 30-minute session.

Use four intervals of width 10. Start at a multiple of 10 that covers all results.

32 18 25 41 29 12 36 22 30 27
15 39 24 44 21 35 19 28 31 26

Interval (X)	Tally	Frequency (Y)
Total		

Draw touching bars. X = sign-ups per session; Y = number of sessions.



How many sessions had at least 30 sign-ups? Show your calculation.

DISTRIBUTIONS AND RELATIONSHIPS

EXERCISE B2

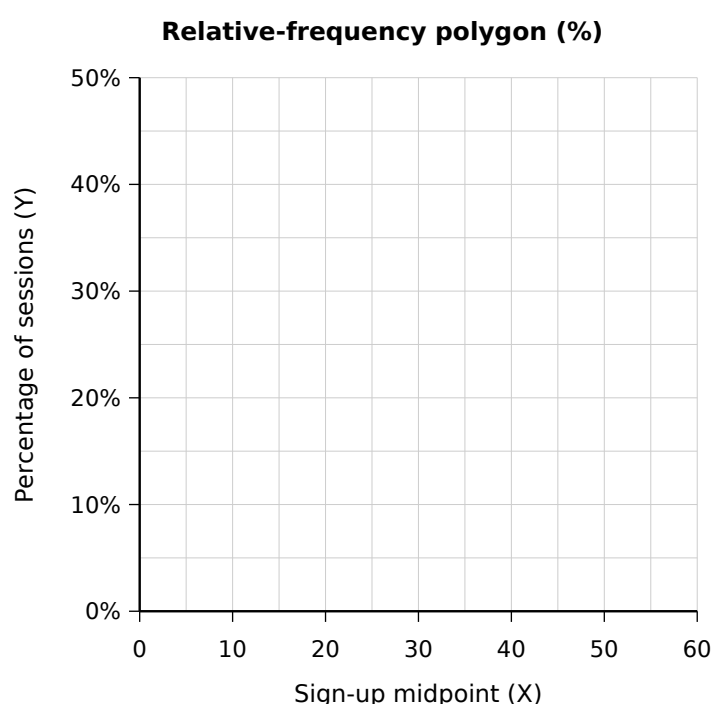
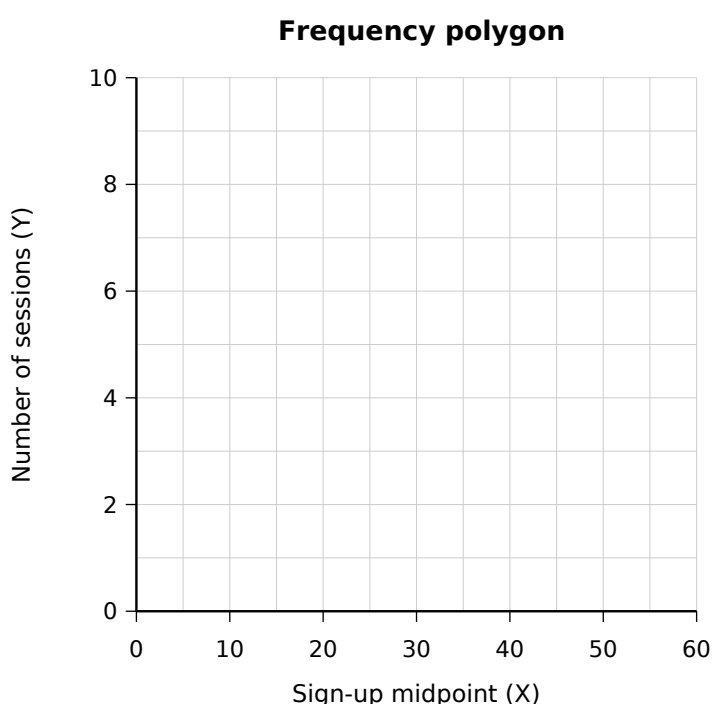
Frequency and relative-frequency polygons

Slides 19-20

Use B1 (20 sessions). Calculate percentages: relative frequencies expressed out of 100.

Midpoint (X)	Frequency (Y)	Percentage (Y)
15	4	
25	8	
35	6	
45	2	

Plot each midpoint and join with straight lines. Close both at (5, 0) and (55, 0).



What changes? What stays the same?

EXERCISE B3

Build the cumulative table

Slides 35-36

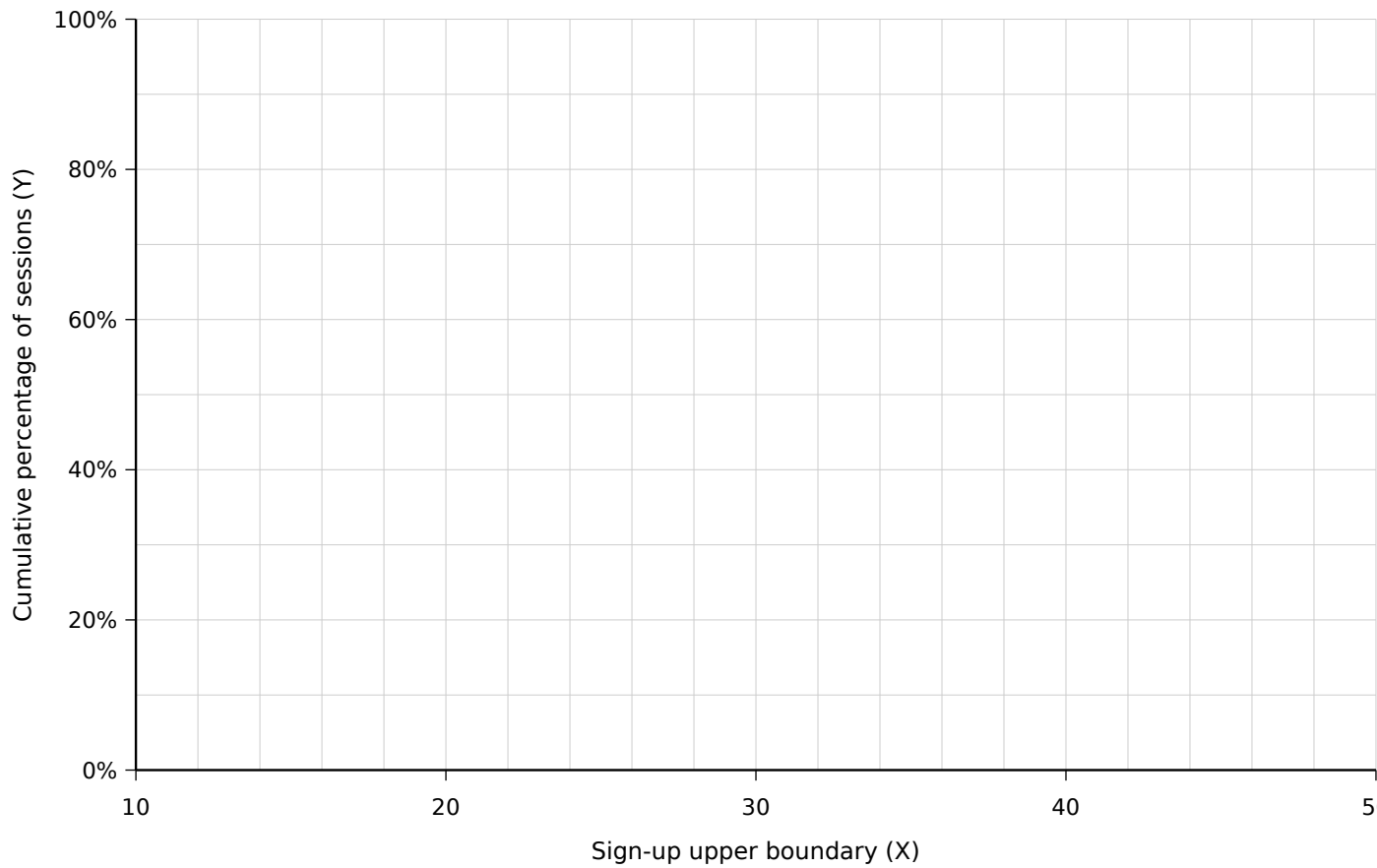
CF is a running total. Divide CF by 20 for cumulative relative frequency.

Interval	Upper boundary (X)	Frequency	CF	Cumulative relative frequency	Cumulative % (Y)
$10 \leq x < 20$	20	4			
$20 \leq x < 30$	30	8			
$30 \leq x < 40$	40	6			
$40 \leq x < 50$	50	2			

DISTRIBUTIONS AND RELATIONSHIPS

EXERCISE B4 Draw the Smoothie Club ogive Slides 37-38

Use the upper boundaries (X) and cumulative percentages (Y) from B3.
Start at (10, 0%). Plot the points and join consecutive points with straight lines.



What percentage of sessions had fewer than 40 sign-ups? _____

Convert this percentage to a number of sessions. Show your working.

EXERCISE B5 Estimate the median from your ogive Slides 39-41

Find 50% on Y. Read across to your ogive, then down to the sign-up axis (X).

1. Which interval contains 50%?	2. Needed rise and full rise (percentage points)
3. Fraction of the rise; distance across the interval	4. Start + distance = estimated median

Why is this an estimate? Discuss what is lost when individual values are grouped.

DISTRIBUTIONS AND RELATIONSHIPS

EXERCISE B6

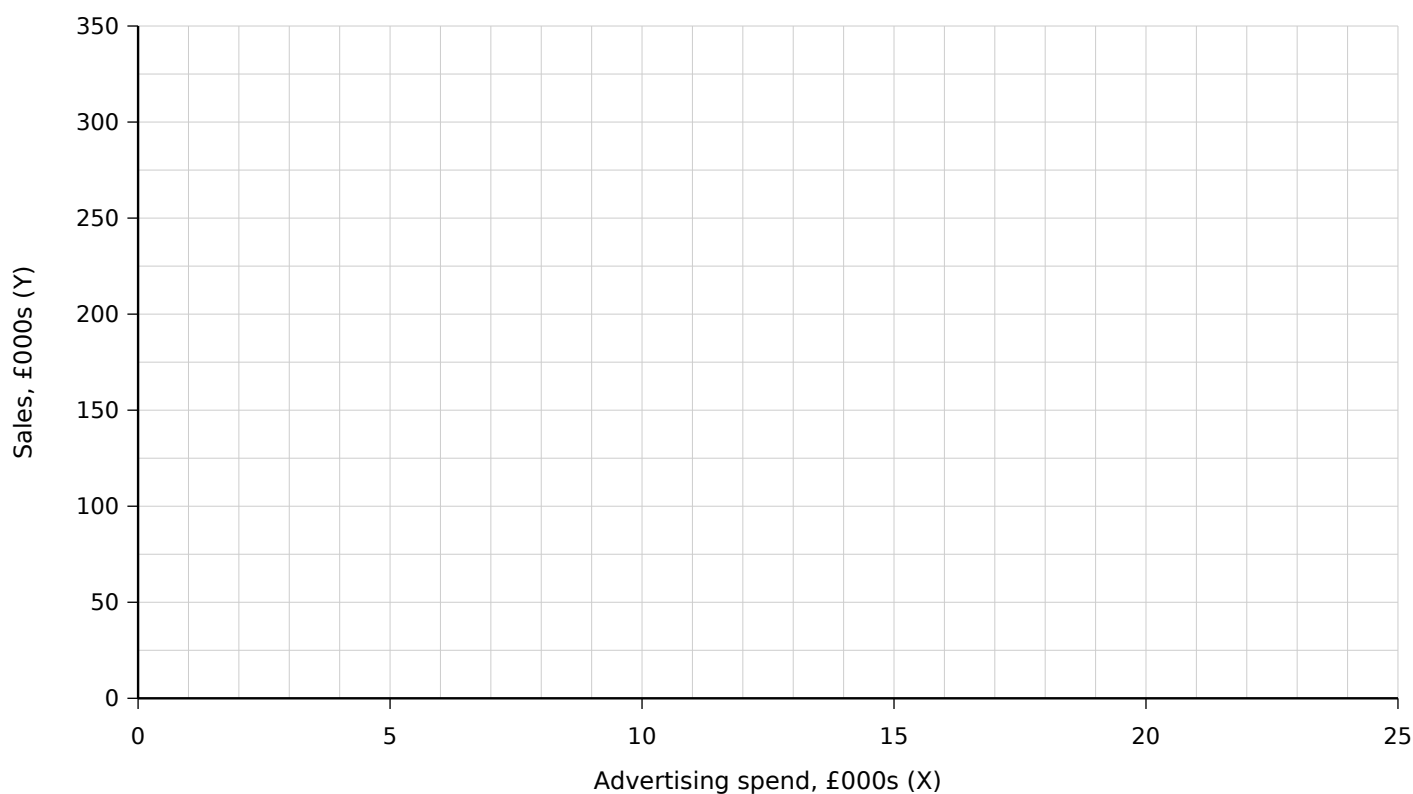
Advertising and sales: plot the pairs

Slides 47-48

Each row is one month. £000s means thousands of pounds: 22.5 = £22,500.

Month	Advertising, £000s (X)	Sales, £000s (Y)
1	15	150
2	22.5	300
3	10.5	120
4	18	180
5	21	225

Plot all five pairs. Keep the two numbers from each month together.



Describe the overall pattern: _____

EXERCISE B7

Use a straight line of best fit

Slides 49-51

Draw one straight line through the centre of the pattern on your scatterplot.

1. Estimate sales when advertising is £20,000. Add dashed reading guides.

Estimated sales: £ _____

2. Would you trust a prediction at £50,000 advertising? Explain why.

DISTRIBUTIONS AND RELATIONSHIPS

EXERCISE B8 Compare two delivery services

Slides 57-58

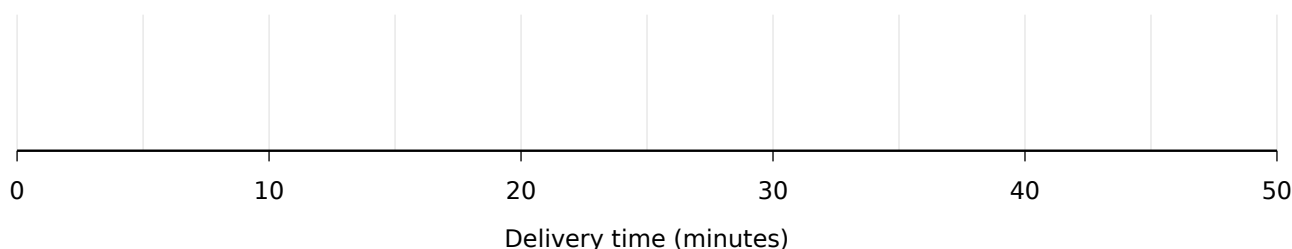
Second service times (minutes): 34, 16, 24, 10, 30, 20, 28, 22, 14, 26, 18.

1. Put the 11 times in ascending order:

2. Calculate the five-number summary. Use the quartile positions from class.

Minimum	Q_1	Median	Q_3	Maximum

3. Draw the boxplot on the 0-50 scale. Label the five numbers.



4. IQR = _____. First service: median 25 min; IQR 14 min. Compare aloud.

EXERCISE B9 Choose a suitable graph

Slides 59-60

Graph bank: bar/pie, histogram, relative-frequency polygon, ogive, scatterplot, boxplot.

Choose a graph for each question. Explain one choice to your partner.

Question	Your graph
Which drinks have the largest shares?	
How many sessions fall in each sign-up interval?	
How do interval shares compare for two sample sizes?	
What percentage had fewer than 30 sign-ups?	
How are advertising and sales related?	
Which service has the lower median and IQR?	

EXERCISE B10 Three final checks

Slides 61-62

1. For $20 \leq x < 30$, which X-value is used? Polygon: ____; ogive: ____.

2. 18 sessions are below 40; 12 are below 30. How many have $30 \leq x < 40$?

3. Does a positive scatterplot prove that X causes Y? Explain aloud.